ThetaSwap: Adverse Competition Oracle

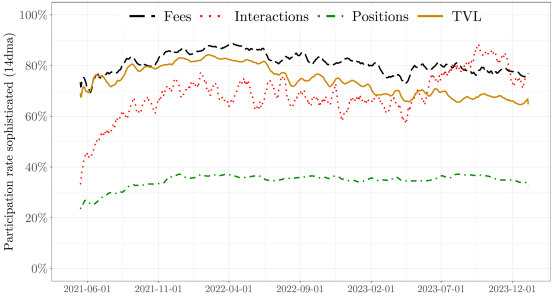
Fee revenue is not equally distributed.

The competition for it is a risk.

Ma and Crapis [2024], Capponi et al. [2024], Aquilina et al. [2024], Milionis et al. [2024]

Approximately 7% of LPs capture around 80% of trading fee revenue

Figure 7: Participation of sophisticated LPs at the protocol level. This figure shows the participation rate of sophisticated market participants across multiple dimensions. Daily pool values are aggregated using a 14 days moving average. Fees is the USD of accrued fees, Interactions is the number of liquidity related transactions, positions is the proportion of fee-collecting liquidity positions and TVL is the total-value locked in USD.



Source: Aquilina et al. [2024], Figure 7

JIT liquidity creates adverse competition —

this is structural, not incidental.

Capponi et al. [2024], Ma and Crapis [2024]

Metrics exist to capture LP competitiveness.

But can you price it? Can you hedge it?

Milionis et al. [2024]

The Oracle: Fee Concentration Index

$$\boxed{\Delta^+ = \max\left(0,\ A_T - A_T^{1/N}\right)}$$

$A_T = \left(\sum_k \theta_k \cdot x_k^2\right)^{1/2}$	fee concentration index
$x_k = L_k / \sum_j L_j$	liquidity share $\approx$ fee share
$\theta_k = 1/\ell_k$	turnover — short-lived positions weigh more
$A_T^{1/N} = \sqrt{\Theta/N^2}$	competitive null (equal shares)

$\Delta^+ > 0$: fees more concentrated than equilibrium.

$\Delta^+ = 0$: fair competition — no adverse regime.

Validation: 600 Real ETH/USDC Positions, 41 Days

Econometric model	
Turning point (protective → adverse)	**9% fee dispersion**
Marginal exit effect above threshold	**+2.3 pp** per %
Willingness to pay for protection	**10% above hourly fees**
Backtest ($200K seed, power-squared payoff)	
Positions better off hedged	**18.8%**
Trigger frequency	**1 day in 41**

Demand exists. It is priceable. The risk profile is textbook insurance.

What's Next

1. Validate across more pools and asset pairs
2. Extend the oracle to other protocols
3. The insurance market — the CFMM to trade this risk on-chain

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